

RADO THEORY FOR JUMP SYSTEMS

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ABSTRACT. We develop a Rado-type theory for nonnegative jump systems. Given a bipartite graph $G = (V_1, V_2, E)$ and a jump system $J_2 \subseteq \mathbb{Z}_+^{V_2}$, we show that J_2 naturally induces a jump system $J_1 \subseteq \mathbb{Z}_+^{V_1}$. This extends the classical theorem of Rado for matroids and its polymatroid generalization due to McDiarmid. Our framework naturally yields several basic operations on jump systems, including identification, splitting, and summation. We then use it to give unified proofs of structural characterizations of discrete polymatroids: constant-sum jump systems are precisely the basis-vector sets of discrete polymatroids, while finite down-closed jump systems are precisely their independent-vector sets. Finally, we derive the König-type theorem and the Rado-type rank function formula for discrete polymatroids from our framework.

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1. INTRODUCTION

Jump systems, introduced by Bouchet and Cunningham [BC95], are subsets of the integer lattice $\mathbb{Z}^V$ for a finite set V that satisfy a two-step axiom. They provide a common framework encompassing several fundamental discrete structures, including matroids, delta-matroids, and discrete polymatroids. In 1935, Hall's famous marriage theorem [Hal35] gave a sufficient (and evidently necessary) condition for a bipartite graph to have a perfect matching. Later, Rado [Rad42] generalized this setting to matroids and proved a well-known theorem showing how a bipartite graph with bipartition (X, Y) and a matroid on Y naturally induces a matroid on X . This was generalized to the setting of integer polymatroids by McDiarmid [McD75]. In this paper, we extend it further to the setting of (nonnegative) jump systems and use it as a unified tool to derive structural results for jump systems, discrete polymatroids, and matroids.

Let V be a finite set and let $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^V$. We shall use the L^1 metric, so we define the *distance* between $\mathbf{x}$ and $\mathbf{y}$ to be

$$\text{dist}(\mathbf{x}, \mathbf{y}) := \sum_{v \in V} |\mathbf{x}_v - \mathbf{y}_v|$$

We say that $\mathbf{x}' \in \mathbb{Z}^V$ is an $(\mathbf{x}, \mathbf{y})$ -step if $\text{dist}(\mathbf{x}, \mathbf{x}') = 1$ and $\text{dist}(\mathbf{x}', \mathbf{y}) = \text{dist}(\mathbf{x}, \mathbf{y}) - 1$.

Definition 1.1. A non-empty set $J \subseteq \mathbb{Z}^V$ is a *jump system* if for all $\mathbf{x}, \mathbf{y} \in J$ and every $(\mathbf{x}, \mathbf{y})$ -step $\mathbf{x}'$,

$$\mathbf{x}' \in J \quad \text{or} \quad \exists \mathbf{x}'' \in J \text{ such that } \mathbf{x}'' \text{ is an } (\mathbf{x}', \mathbf{y})\text{-step.}$$

We call this condition *two-step axiom* and V the *ground set* of J . Here are some fundamental examples of jump systems that we shall keep in mind:

Example 1.2. Let M be a matroid on ground set E . The collection $\mathbf{I}_M := \{\mathbf{1}_I \mid I \in \mathcal{I}_M\} \subseteq \mathbb{Z}_+^E$ is a jump system. To see this, call $\mathbf{x}$ *valid* if $\mathbf{x} \in \mathbf{I}_M$. Take any $\mathbf{1}_I, \mathbf{1}_J \in \mathbf{I}_M$. Then, an $(\mathbf{1}_I, \mathbf{1}_J)$ -step is a vector $\mathbf{t}$ indicating either $I + e$ where $e \in J \setminus I$, or $I - e$ where $e \in I \setminus J$. If we are in the second case, then since $I - e \in \mathcal{I}_M$ by the hereditary property of independent sets, $\mathbf{t}$ is valid and we are done. If we are in the first case, then we have two possibilities:

- α . $I + e \in \mathcal{I}_M$. Then $\mathbf{t}$ is valid and we are done.
- β . $I + e \notin \mathcal{I}_M$. This implies that we have a unique circuit $C \subseteq I + e$ containing e . Thus, there exists an element $f \in I \setminus J$ such that $I + e - f \in \mathcal{I}_M$. Let $\mathbf{t}' := \mathbf{1}_{I+e-f}$. Then $\mathbf{t}'$ is valid and is an $(\mathbf{t}, \mathbf{1}_J)$ -step, so we are done.

Definition 1.3. A jump system $J \subseteq \mathbb{Z}^V$ is *constant-sum* if there exists $c \in \mathbb{Z}$ such that

$$\sum_{v \in V} \mathbf{x}_v = c \text{ for all } \mathbf{x} \in J.$$

We call c the *coordinate sum*¹ of J .

Example 1.4. Let M be a matroid on ground set E . The collection $\mathbf{B}_M := \{\mathbf{1}_B \mid B \in \mathcal{B}_M\} \subseteq \mathbb{Z}_+^E$ is a constant-sum jump system. The proof is similar to that of Example 1.2, using the basis exchange property [Oxl11, Lemma 1.2.2]. Note that all members of $\mathcal{B}_M$ have the same cardinality, so $\mathbf{B}_M$ is constant-sum.

Example 1.5. Fix an arbitrary graph $G = (V, E)$. Let $\mathbf{d}^F$ denote the degree vector of the spanning subgraph (V, F) . That is

$$\mathbf{d}_v^F = |\{e \in F : e \text{ is incident with } v\}| \quad \text{for every } v \in V$$

We call $\mathbf{D}_G := \{\mathbf{d}^F : F \subseteq E\} \subseteq \mathbb{Z}_+^V$ the *degree-sequence system* of G . It follows directly from Corollary 4.7 that $\mathbf{D}_G$ is a jump system.

Remark 1.6. Although they are not our concern in this paper, many other specific sets are known as jump systems. Here we mention a few. Brändén [Brä07] showed that the support of a multivariate polynomial with the half-plane property is a jump system. Kobayashi and Takazawa [KT09] showed that the degree sequences of all even factors in a digraph form a jump system if and only if the digraph is odd-cycle-symmetric.

Important assumption: In the rest of the paper, unless explicitly specified, we assume that our jump systems are all finite. We now introduce our Rado theory for jump systems.

Definition 1.7. If $G = (V, E)$ is a graph and $\mathbf{f} \in \mathbb{Z}_+^E$, we define the *boundary vector* $\partial \mathbf{f} \in \mathbb{Z}_+^V$ by the rule

$$(\partial \mathbf{f})_v := \sum_{e \sim v} \mathbf{f}_e$$

where $v \in V$ and $e \sim v$ means that $e \in E$ is incident with v .

It will be helpful to consider bipartite graphs that are equipped with a distinguished bipartition. In this case we write $G = (V_1, V_2, E)$ to indicate that (V_1, V_2) is the distinguished bipartition of $V := V(G)$. We frequently need to consider the restriction of our boundary function to one side of the bipartition, so for every $\mathbf{f} \in \mathbb{Z}_+^E$ and $i = 1, 2$ we define $\partial_i \mathbf{f}$ to be the restriction of $\partial \mathbf{f}$ to V_i . Now, we state our first main result as follows:

¹In general, for any vector $\mathbf{x} \in \mathbb{Z}^V$, we call $\sum_{v \in V} \mathbf{x}_v$ the *coordinate sum* of $\mathbf{x}$.

Theorem 1.8 (Theorem 4.3). *If $G = (V_1, V_2, E)$ is a bipartite graph and $J_2 \subseteq \mathbb{Z}_+^{V_2}$ is a jump system, then $J_1 := \{\partial_1 \mathbf{f} \mid \mathbf{f} \in \mathbb{Z}_+^E \text{ satisfies } \partial_2 \mathbf{f} \in J_2\}$ is also a jump system.*

Remark 1.9. In the setting above, we call the jump system $J_1 \subseteq \mathbb{Z}_+^{V_1}$ given by Theorem 4.3 the jump system *associated* to J_2 via G , denoted as $\Theta_G(J_2)$. There are two specially structured bipartite graphs for which these associated jump systems deserve special names: If the graph G satisfies $\deg(v) = 1$ for every $v \in V_2$ (respectively, every $v \in V_1$), then we call J_1 an *identification* (respectively, a *splitting*) of J_2 .

Fix an arbitrary bipartite graph $G = (V_1, V_2, E)$. The classic Rado theorem for matroids can be stated as follows:

Theorem 1.10 (Rado). *Given a matroid M_2 on V_2 . M_2 naturally induces a matroid M_1 on V_1 via G that can be characterized by the following:*

- A. $X \in \mathcal{I}_{M_1} \iff X$ can be matched to a set $Y \in \mathcal{I}_{M_2}$ via G
 $\iff |X'| \leq \text{rk}_{M_2}(N(X'))$ for every $X' \subseteq X$
- B. $\text{rk}_{M_1}(S) = \min_{T \subseteq S} \{\text{rk}_{M_2}(N(T)) + |S \setminus T|\}$ for every subset $S \subseteq V_1$.

Remark 1.11. It is evident that, in our language, if we encode the matroid M_2 by its independent vectors $\mathbf{I}_{M_2}$ as a jump system, then

$$\mathbf{I}_{M_1} = \Theta_G(\mathbf{I}_{M_2}) \cap [\mathbf{0}, \mathbf{e}]$$

Polymatroids, introduced by Edmonds [Edm70], generalize matroids. And just like matroids, polymatroids have several equivalent definitions. In this paper, we focus on discrete polymatroids, which can be defined as follows:

Definition 1.12. A *discrete polymatroid* P on the ground set E is a function $\text{rk}_P : 2^E \rightarrow \mathbb{Z}_+$ satisfying the following properties:

- PM1. $\text{rk}_P(\emptyset) = 0$.
- PM2. $\text{rk}_P(S') \leq \text{rk}_P(S)$ whenever $S' \subseteq S \subseteq E$.
- PM3. $\text{rk}_P(S) + \text{rk}_P(T) \geq \text{rk}_P(S \cap T) + \text{rk}_P(S \cup T)$, for all $S, T \in 2^E$.

where PM1, PM2, and PM3 are called *normalization*, *monotonicity*, and *submodularity*, respectively. As in the case of matroids, we call rk_P the *rank function* of P .

Note that given a matroid M on V with rank function rk_M , we may interpret the sets $\mathbf{I}_M$ and $\mathbf{B}_M$ defined in the previous examples (1.2 and 1.4) as follows:

$$\begin{aligned} \mathbf{I}_M &= \{\mathbf{x} \in \mathbb{Z}_+^V \mid \mathbf{x}_S \leq \text{rk}_M(S) \text{ for every } S \subseteq V\} \\ \mathbf{B}_M &= \{\mathbf{x} \in \mathbb{Z}_+^V \mid \mathbf{x}_S \leq \text{rk}_M(S) \text{ for every } S \subseteq V \text{ and } \mathbf{x}_V = \text{rk}_M(V)\} \end{aligned}$$

Similarly, for a given integer polymatroid P on V' , we define the following collections of lattice points:

$$\begin{aligned} \mathbf{I}_P &= \{\mathbf{x} \in \mathbb{Z}_+^{V'} \mid \mathbf{x}_S \leq \text{rk}_P(S) \text{ for every } S \subseteq V'\} \\ \mathbf{B}_P &= \{\mathbf{x} \in \mathbb{Z}_+^{V'} \mid \mathbf{x}_S \leq \text{rk}_P(S) \text{ for every } S \subseteq V' \text{ and } \mathbf{x}_{V'} = \text{rk}_P(V')\} \end{aligned}$$

We call the elements in $\mathbf{I}_M$ and $\mathbf{I}_P$ *independent vectors*; elements in $\mathbf{B}_M$ and $\mathbf{B}_P$ *basis vectors*. Note that we can recover the rank function of P by $\text{rk}_P(S) = \max\{\mathbf{x}_S : \mathbf{x} \in \mathbf{I}_P\}$ for every $S \subseteq V'$. Now, we state our second main theorem as follows:

Theorem 1.13 (Theorem 5.2). *For every $J \subseteq \mathbb{Z}_+^V$, the following are equivalent:*

- (i). $J = \mathbf{B}_P$ for a discrete polymatroid P .

- (ii). J is constant-sum and for every $\mathbf{x}, \mathbf{y} \in J$ and $i \in V$ with $\mathbf{x}_i < \mathbf{y}_i$, there exists $j \in V$ with $\mathbf{x}_j > \mathbf{y}_j$ so that both $\mathbf{x} + \mathbf{e}_i - \mathbf{e}_j \in J$ and $\mathbf{y} - \mathbf{e}_i + \mathbf{e}_j \in J$.
- (iii). J is a constant-sum jump system.
- (iv). J can be obtained from $\mathbf{B}_M$ for a matroid M by identification.

Remark 1.14. We shall emphasize that we do not claim significant novelty for Theorem 5.2. To the best of our knowledge, (i) $\iff$ (ii) was proved by Herzog and Hibi [HH02]; (i) $\iff$ (iv) was first realized by Helgason [Hel74]; (iii) $\iff$ (iv) was essentially due to Kabadi and Sridhar [KS05]. However, we believe that we are the first to put the above equivalent statements together, and we will give a clean proof by showing (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (iv) $\implies$ (i) using our language developed earlier. In particular, our proof of (i) $\implies$ (ii) largely simplifies the argument in [HH02].

Definition 1.15. Let $K \subseteq \mathbb{Z}_+^V$ be a set of vectors. We define its *down-closure* by

$$\downarrow K := \{\mathbf{x} \in \mathbb{Z}_+^V : \mathbf{x} \preceq \mathbf{y} \text{ for some } \mathbf{y} \in K\}$$

We say that K is *down-closed* if $\downarrow K = K$.

As a consequence of Theorem 5.2, we recover the following independent vector version of the characterization: finite down-closed jump systems are precisely the independent vector sets of discrete polymatroids.

Theorem 1.16 (Theorem 5.4). *For every $J \subseteq \mathbb{Z}_+^V$, the following are equivalent:*

- (i). $J = \mathbf{I}_P$ for a discrete polymatroid P .
- (ii). J is a down-closed jump system.
- (iii). J can be obtained from $\mathbf{I}_M$ for a matroid M by identification.

Edmonds' celebrated matroid intersection theorem [Edm70] gives a min-max characterization of the maximum cardinality of a common independent set of two matroids on the same ground set. It can be stated as follows.

Theorem 1.17 (Matroid Intersection, Edmonds). *Let M_1 and M_2 be matroids on the same ground set S . Then*

$$\max_{I \in \mathcal{I}_{M_1} \cap \mathcal{I}_{M_2}} |I| = \min_{X \subseteq S} \text{rk}_{M_1}(X) + \text{rk}_{M_2}(S \setminus X)$$

A corresponding polymatroid intersection theorem, already implicit in Edmonds' work, was later stated explicitly by McDiarmid [McD75]. Our framework naturally yields the following discrete version.

Theorem 1.18 (König-type Theorem for Discrete Polymatroids 6.2). *Let $G = (V_1, V_2, E)$ be a bipartite graph, and let P_1, P_2 be discrete polymatroids on V_1, V_2 , respectively. Then*

$$\max_{\substack{\partial_1 \mathbf{f} \in \mathbf{I}_{P_1} \\ \partial_2 \mathbf{f} \in \mathbf{I}_{P_2}}} \{\mathbf{f}_E\} = \min_{\substack{X_1 \subseteq V_1 \\ X_2 \subseteq V_2 \\ X_1 \sqcup X_2 \text{ is a vertex cover}}} \{\text{rk}_{P_1}(X_1) + \text{rk}_{P_2}(X_2)\}$$

Definition 1.19. Following Eur and Larson [EL24], we say that a discrete polymatroid P on V with a *cage*² $\mathbf{c} \in \mathbb{Z}_+^V$ is a discrete polymatroid satisfying an extra property:

PM4. $\text{rk}_P(\{v\}) \leq \mathbf{c}_v$ for every $v \in V$.

Remark 1.20. A matroid is exactly a polymatroid with a cage $\mathbf{e}$.

²The terminology ‘‘cage’’ is due to Eur and Larson [EL24], while the use of a coordinatewise bounding vector for polymatroids goes back at least to McDiarmid [McD75].

Theorem 1.21 (Rado Rank Formula for Discrete Polymatroids 6.3). *Let $G = (V_1, V_2, E)$ be a bipartite graph, P_2 be a discrete polymatroid on V_2 , and let $\mathbf{c} \in \mathbb{Z}_+^{V_1}$ be a cage. Then*

$$\{\mathbf{x} \in \mathbb{Z}_+^{V_1} : \mathbf{x} \preceq \mathbf{c}, \mathbf{x} = \partial_1 \mathbf{f}, \partial_2 \mathbf{f} \in \mathbf{I}_{P_2} \text{ for some } \mathbf{f} \in \mathbb{Z}_+^E\}$$

is the set of independent vectors $\mathbf{I}_{P_1}$ of a discrete polymatroid P_1 on V_1 , with cage $\mathbf{c}$. Moreover, the rank function of P_1 is given by

$$\text{rk}_{P_1}(S) = \min_{T \subseteq S} \{\text{rk}_{P_2}(N(T)) + \mathbf{c}_{S \setminus T}\}$$

for every $S \subseteq V_1$.

Remark 1.22. If we take the cage $\mathbf{c} = \mathbf{e}$, then the above theorem reduces to assertion B of Theorem 1.10, in view of Remark 1.11.

This paper is organized as follows. Section 2 fixes our conventions and notation, and Section 3 collects several basic facts about jump systems that will be used throughout the paper. In Section 4, we prove our main Rado-type theorem for jump systems and derive a number of structural consequences, including identification, splitting, and summation operations. Section 5 develops the connection between jump systems and discrete polymatroids, giving characterizations of both basis-vector and independent-vector systems. Finally, in Section 6, we use this framework to obtain the König-type theorem and the Rado-type rank function formula for discrete polymatroids.

2. CONVENTIONS AND NOTATIONS

- (1) Let V be a finite set and $\mathbb{Z}^V$ be the free abelian group generated by V . We use $\mathbb{Z}_+$ to denote the non-negative integers and $\mathbb{Z}_+^V$ to denote the truncation of $\mathbb{Z}^V$. Elements in $\mathbb{Z}^V$ will be denoted by bold lowercase letters like $\mathbf{f}, \mathbf{g}, \mathbf{x}, \mathbf{y}$, which we call *vectors*. In particular, $\mathbf{e}$ will denote the vector whose entries are all 1 and $\mathbf{0}$ will denote the vector whose entries are all 0. Let $A \subseteq V$. We say $\mathbf{1}_A$ is the *indicating vector* of A where

$$(\mathbf{1}_A)_v = \begin{cases} 1 & \text{if } v \in A \\ 0 & \text{if } v \notin A \end{cases}$$

For any subset $S \subseteq V$, let $\mathbf{f}_S := \sum_{s \in S} \mathbf{f}_s$ where $\mathbf{f}_s$ is defined to be the value of $\mathbf{f}$ on $s \in V$. For $\mathbf{x} = (x_1, \dots, x_m) \in \mathbb{Z}^{V_1}$ and $\mathbf{y} = (y_1, \dots, y_n) \in \mathbb{Z}^{V_2}$ where $|V_1| = m, |V_2| = n$, and $V_1 \cap V_2 = \emptyset$, define $\mathbf{x} \oplus \mathbf{y} := (x_1, \dots, x_m, y_1, \dots, y_n) \in \mathbb{Z}^{V_1 \sqcup V_2}$. For $\mathbf{x}, \mathbf{y} \in \mathbb{Z}_+^V$, we say $\mathbf{x} \preceq \mathbf{y}$ if $\mathbf{x}_v \leq \mathbf{y}_v$ for all $v \in V$.

- (2) For any subset $A, B \subseteq \mathbb{Z}^V$ and any vector $\mathbf{x} \in \mathbb{Z}^V$, we define the *distance* between $\mathbf{x}$ and A to be

$$\text{dist}(\mathbf{x}, A) := \min \{\text{dist}(\mathbf{x}, \mathbf{z}) \mid \mathbf{z} \in A\}$$

and define the *distance* between A and B to be

$$\text{dist}(A, B) := \min \{\text{dist}(\mathbf{z}, B) \mid \mathbf{z} \in A\}$$

- (3) In a graph, a *walk* is a sequence of edges which joins a sequence of vertices; a *trail* is a walk in which all edges are distinct. Let $G = (V, E)$ be a graph. For any $U \subseteq V$, we define its *neighbor* as the set of vertices that are adjacent to some $u \in U$, denoted as $N(U)$.
- (4) Given a matroid M , we use $\mathcal{I}_M$ to denote the collection of independent sets of M and use $\mathcal{B}_M$ to denote the collection of bases of M .
- (5) Given a set E , let $A \subseteq E$ and $e \in E$. We write $A + e := A \cup \{e\}$ and $A - e := A \setminus \{e\}$.

3. SOME DEFINITIONS AND BASIC FACTS

Definition 3.1. We define an analogue of the direct sum of matroids for jump systems. Let $J_1, \dots, J_n$ be jump systems on pairwise disjoint ground sets $X_1, \dots, X_n$. Then the set

$$J := \{\mathbf{x}_1 \oplus \dots \oplus \mathbf{x}_n \mid \mathbf{x}_i \in J_i, \forall i \in [n]\}$$

is clearly a jump system, called the *direct sum* of $J_1, \dots, J_n$ and denoted by $\bigoplus_{i=1}^n J_i$.

Proposition 3.2. For any vectors $\mathbf{x}, \mathbf{x}'$, and $\mathbf{y} \in \mathbb{Z}^V$ satisfying $\text{dist}(\mathbf{x}, \mathbf{x}') = 1$, either $\mathbf{x}'$ is an $(\mathbf{x}, \mathbf{y})$ -step, or $\mathbf{x}$ is an $(\mathbf{x}', \mathbf{y})$ -step.

Proof. Since $\text{dist}(\mathbf{x}, \mathbf{x}') = 1$, we know that $\mathbf{x}_v = \mathbf{x}'_v$ for all $v \in V$ but one, say w , where $|\mathbf{x}_w - \mathbf{x}'_w| = 1$. Suppose $\mathbf{x}'$ is not an $(\mathbf{x}, \mathbf{y})$ -step. Then $\text{dist}(\mathbf{x}', \mathbf{y}) \geq \text{dist}(\mathbf{x}, \mathbf{y})$. Thus,

$$\sum_{v \in V} |\mathbf{y}_v - \mathbf{x}'_v| = \sum_{v \neq w} |\mathbf{y}_v - \mathbf{x}'_v| + |\mathbf{y}_w - \mathbf{x}'_w| \geq \sum_{v \neq w} |\mathbf{y}_v - \mathbf{x}_v| + |\mathbf{y}_w - \mathbf{x}_w| = \sum_{v \in V} |\mathbf{y}_v - \mathbf{x}_v|$$

Therefore, $|\mathbf{y}_w - \mathbf{x}'_w| \geq |\mathbf{y}_w - \mathbf{x}_w|$. Hence $|\mathbf{y}_w - \mathbf{x}'_w| = |\mathbf{y}_w - \mathbf{x}_w| + 1$. But this implies that $\mathbf{x}$ is an $(\mathbf{x}', \mathbf{y})$ -step, as desired. The other direction follows by symmetry. $\square$

Definition 3.3. Let $V = [n]$. For $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^V$, define

$$[\mathbf{x}, \mathbf{y}] := \{\mathbf{z} \in \mathbb{Z}^V \mid \min\{\mathbf{x}_i, \mathbf{y}_i\} \leq \mathbf{z}_i \leq \max\{\mathbf{x}_i, \mathbf{y}_i\}\}$$

We call $[\mathbf{x}, \mathbf{y}]$ a *box*. In other words,

$$[\mathbf{x}, \mathbf{y}] = [\mathbf{x}_1, \mathbf{y}_1] \times \dots \times [\mathbf{x}_n, \mathbf{y}_n]$$

where $[\mathbf{x}_i, \mathbf{y}_i]$ are intervals. Let $B = [\mathbf{x}, \mathbf{y}]$ and let $J \subseteq \mathbb{Z}^V$ be a jump system. We say a vector $\mathbf{z} \in J$ is *optimal* with respect to B if

$$\text{dist}(\mathbf{z}, B) = \text{dist}(J, B)$$

The collection of all such optimal vectors is denoted by J_B .

Proposition 3.4 (Lovász[Lov97]). Let $J \subseteq \mathbb{Z}_+^V$ be a jump system and $B \subseteq \mathbb{Z}_+^V$ be a box. Then J_B is a jump system.

Remark 3.5. A special case of Proposition 3.4 is when $J \cap B \neq \emptyset$. Then $J_B = J \cap B$.

Proposition 3.6. Let $J \subseteq \mathbb{Z}_+^V$ be a constant-sum jump system. Then $J \cap [0, \mathbf{e}]$, if not empty, is equal to $\mathbf{B}_M$ for some matroid M .

Proof. Let $\mathbf{B} := J \cap [0, \mathbf{e}]$. Identify $\mathbf{B}$ with a family of subsets of V , denoted as $\mathcal{B}$. Since $\mathbf{B}$ is constant-sum, all sets in $\mathcal{B}$ have the same cardinality. It remains to check the basis exchange axiom. Take $A, B \in \mathcal{B}$, and let $a \in A \setminus B$. Then $\mathbf{1}_A - \mathbf{e}_a$ is an $(\mathbf{1}_A, \mathbf{1}_B)$ -step. Since $\mathbf{B}$ is constant-sum, this vector is not in $\mathbf{B}$. Hence, the two-step axiom gives another vector of $\mathbf{B}$ obtained from $\mathbf{1}_A - \mathbf{e}_a$ by taking one more step towards $\mathbf{1}_B$. The only possible such step is to add some element $b \in B \setminus A$. Therefore

$$\mathbf{1}_A - \mathbf{e}_a + \mathbf{e}_b \in \mathbf{B}$$

That is

$$A - a + b \in \mathcal{B}$$

as desired. Hence $\mathcal{B} = \mathcal{B}_M$ for some matroid M on V and $\mathbf{B} = \mathbf{B}_M$. $\square$

Proposition 3.7. Let $J \subseteq \mathbb{Z}_+^V$ be a jump system, let $m = \min\{\mathbf{x}_V \mid \mathbf{x} \in J\}$ and let $M = \max\{\mathbf{x}_V \mid \mathbf{x} \in J\}$. Then the following sets are constant-sum jump systems.

$$J_{\min} := \{\mathbf{x} \in J \mid \mathbf{x}_V = m\} \quad J_{\max} := \{\mathbf{x} \in J \mid \mathbf{x}_V = M\}$$

Proof. It follows directly from Proposition 3.4 by noting that a box can be a single vector. Observe that $J_{\min} = J_0$ and $J_{\max} = J_u$ where $u_v = \max\{x_v \mid x \in J\}$. $\square$

Proposition 3.8. *Let $J \subseteq \mathbb{Z}_+^V$ be a jump system. Then, for any $x \in J$, there exists $y \in J_{\max}$ such that $x \preceq y$.*

Proof. Denote V as $[n]$ since it is finite. Fix an arbitrary $x \in J$. Among all the vectors in $J_{\max}$ choose y such that $\text{dist}(x, y)$ is minimized. We claim that $x \preceq y$. Suppose the contrary. Then there exists an index $i \in [n]$ such that $x_i > y_i$ hence $y + e_i$ is an (y, x) -step. By assumption $y + e_i \notin J$ and $y + e_i + e_j \notin J$ for every $j \in [n]$. Hence, by the two-step axiom, there exists an index $j \in [n]$ such that $y + e_i - e_j$ is in J and is an $(y + e_i, x)$ -step. This implies that $y + e_i - e_j$ is also in $J_{\max}$ but $\text{dist}(y + e_i - e_j, x) < \text{dist}(y, x)$, contradicting our choice of y . $\square$

4. RADO THEORY FOR JUMP SYSTEMS

Definition 4.1. A jump system $J \subseteq \mathbb{Z}^V$ is called an *even jump system* if

$$\text{dist}(x, y) \text{ is even for all } x, y \in J.$$

In other words, a non-negative collection of vectors $J \subseteq \mathbb{Z}^V$ is an even jump system if and only if $\forall x, y \in J$ and $\forall (x, y)$ -step x' , we have $x' \notin J$ but $\exists (x', y)$ -step x'' such that $x'' \in J$. Note that constant-sum jump systems are clearly even. The next lemma shows that for any loopless graph, the collection of all boundary vectors forms an even jump system.

Lemma 4.2. *For every loopless graph $G = (V, E)$ the set $J := \{\partial f \mid f \in \mathbb{Z}_+^E\}$ is an even jump system.*

Proof. First, we note that for any $f \in \mathbb{Z}_+^E$, the boundary vector ∂f has even coordinate sum, so $\text{dist}(x, y)$ is even for all $x, y \in J$. Therefore, if J is a jump system, it has to be even. Thus, it suffices to show the following:

$$\forall x, y \in J \text{ and } \forall (x, y)\text{-step } x', \exists x'' \in J \text{ such that } x'' \text{ is an } (x', y)\text{-step}.$$

In fact, we may reduce the problem to showing that $J^\natural := \{\partial f \mid f \in \{0, 1\}^E\}$ is an even jump system, because once we have done this, we could recover J by allowing parallel edges on G .

Now, fix any $x, y \in J^\natural$. If $x = y$ then there is nothing to prove, so we may assume $x \neq y$. Let $f, h \in \{0, 1\}^E$ satisfy $x = \partial f, y = \partial h$. Set $m \in \{0, 1\}^E$ where $m_e := \min\{f_e, h_e\}$ for all $e \in E$ and define

$$\tilde{f} := f - m \quad \text{and} \quad \tilde{h} := h - m$$

We call this operation *reduction*. Note that this is also a translation in $\{0, 1\}^V$, namely

$$\partial \tilde{f} = \partial f - \partial m \quad \text{and} \quad \partial \tilde{h} = \partial h - \partial m$$

That is, if we obtain a desired vector $\tilde{g}$ in the reduced case, we could set $g := \tilde{g} + m$ so that $\partial g = \partial(\tilde{g} + m) \in J^\natural$. One might worry that $g \notin \{0, 1\}^E$; however, note that $\text{Supp}(m) \cap (\text{Supp}(\tilde{f}) \cup \text{Supp}(\tilde{h})) = \emptyset$, our argument later will show that $\text{Supp}(\tilde{g}) \subseteq \text{Supp}(\tilde{f}) \cup \text{Supp}(\tilde{h})$, hence $\text{Supp}(g) \cap \text{Supp}(m) = \emptyset$, so $\tilde{g} + m \in \{0, 1\}^E$, as required.

Therefore, we may assume that f and h are reduced. Now, since for each $e \in E$, f_e and h_e cannot both be 1, we can define a proper coloring as follows:

$$\text{For each edge } e \in E, \text{ color } e \begin{cases} \text{red} & \text{if } f_e = 1 \\ \text{blue} & \text{if } h_e = 1 \end{cases}$$

Discard the uncolored edges. For each vertex $w \in V$, let $\deg_{\text{red}}(w)$ denote the number of red edges incident with w , $\deg_{\text{blue}}(w)$ denote the number of blue edges incident with w . We say that w is *dominated by red*

if $\deg_{\text{red}}(w) > \deg_{\text{blue}}(w)$, that w is *dominated by blue* if $\deg_{\text{red}}(w) < \deg_{\text{blue}}(w)$, and that w is *balanced* if $\deg_{\text{red}}(w) = \deg_{\text{blue}}(w)$. Let $\mathbf{x}'$ be an arbitrary $(\mathbf{x}, \mathbf{y})$ -step. Then there exists a unique $v \in V$ such that $|\mathbf{x}_v - \mathbf{x}'_v| = 1$. Note that v cannot be balanced, because if so, then $\mathbf{x}_v = \mathbf{y}_v$. Changing the v -coordinate by ± 1 would increase the distance to $\mathbf{y}$, contradicting our assumption that $\mathbf{x}'$ is an $(\mathbf{x}, \mathbf{y})$ -step. Therefore, we have two possible cases:

α . v is dominated by red, i.e., $\deg_{\text{red}}(v) > \deg_{\text{blue}}(v)$. So $\mathbf{x}' = \mathbf{x} - \mathbf{e}_v$. Now, starting from v with an arbitrary red edge, find a maximal alternating trail $\mathcal{T}$ (we may psychologically delete an edge from G once we put it in $\mathcal{T}$), denote the end vertex by u . We claim that $\mathcal{T}$ has exactly one of the following two endpoint types:

$$\text{I: } v \xrightarrow{\text{red}} \bullet \cdots \bullet \xrightarrow{\text{red}} \bullet u \quad \text{or} \quad \text{II: } v \xrightarrow{\text{red}} \bullet \cdots \bullet \xrightarrow{\text{blue}} \bullet u$$

In case I, u is dominated by red and e_{end} is red; in case II, u is dominated by blue and e_{end} is blue. That is, the dominant color of u and the color of e_{end} must be the same. To see this, suppose otherwise that e_{end} is colored c_1 and u is dominated by c_2 where $c_1 \neq c_2$, then there exists an edge $e \in E$ colored c_2 and incident with u such that $e \notin \mathcal{T}$, contradicting the maximality of $\mathcal{T}$. Now, switch the color of every edge $e \in \mathcal{T}$ and set $\mathbf{g} \in \{0, 1\}^E$ where

$$\text{For each edge } e \in E, \mathbf{g}_e = \begin{cases} 1 & \text{if } e \text{ is colored red} \\ 0 & \text{if } e \text{ is colored blue} \end{cases}$$

So $\partial \mathbf{g} \in \mathbb{J}^\natural$. If we are in case I, then $\partial \mathbf{g} = \mathbf{x} - \mathbf{e}_u - \mathbf{e}_v$; if we are in case II, then $\partial \mathbf{g} = \mathbf{x} + \mathbf{e}_u - \mathbf{e}_v$. Note that in both cases, $\partial \mathbf{g}$ is an $(\mathbf{x}', \mathbf{y})$ -step, so we are done.

β . v is dominated by blue, i.e., $\deg_{\text{red}}(v) < \deg_{\text{blue}}(v)$. So $\mathbf{x}' = \mathbf{x} + \mathbf{e}_v$. By the same reasoning as above, any maximal trail $\mathcal{T}$ starting from v with a blue edge will be one of the following two endpoint types:

$$\text{III: } v \xrightarrow{\text{blue}} \bullet \cdots \bullet \xrightarrow{\text{blue}} \bullet u \quad \text{or} \quad \text{IV: } v \xrightarrow{\text{blue}} \bullet \cdots \bullet \xrightarrow{\text{red}} \bullet u$$

Again, switch the color of every edge $e \in \mathcal{T}$ and set $\mathbf{g} \in \{0, 1\}^E$ by the same rule as above. If we are in case III, then $\partial \mathbf{g} = \mathbf{x} + \mathbf{e}_u + \mathbf{e}_v$; if we are in case IV, then $\partial \mathbf{g} = \mathbf{x} - \mathbf{e}_u + \mathbf{e}_v$. In both cases, $\partial \mathbf{g}$ is an $(\mathbf{x}', \mathbf{y})$ -step, so we are done. $\square$

Theorem 4.3. *If $G = (V_1, V_2, E)$ is a bipartite graph and $\mathbb{J}_2 \subseteq \mathbb{Z}_+^{V_2}$ is a jump system, then $\mathbb{J}_1 := \{\partial_1 \mathbf{f} \mid \mathbf{f} \in \mathbb{Z}_+^E \text{ satisfies } \partial_2 \mathbf{f} \in \mathbb{J}_2\}$ is also a jump system.*

Proof. Call $\mathbf{f} \in \mathbb{Z}_+^E$ valid if $\partial_2 \mathbf{f} \in \mathbb{J}_2$. To prove that $\mathbb{J}_1$ is a jump system, let $\mathbf{x}, \mathbf{y} \in \mathbb{J}_1$ and let $\mathbf{x}'$ be an arbitrary $(\mathbf{x}, \mathbf{y})$ -step. We need to show that $\mathbf{x}' \in \mathbb{J}_1$, or there exists an $(\mathbf{x}', \mathbf{y})$ -step $\mathbf{x}'' \in \mathbb{J}_1$. By assumption, we may choose a valid $\mathbf{h} \in \mathbb{Z}_+^E$ so that $\partial_1 \mathbf{h} = \mathbf{y}$. Note that there exists valid $\mathbf{f} \in \mathbb{Z}_+^E$ such that $\text{dist}(\partial_1 \mathbf{f}, \mathbf{x}') = 1$ since $\text{dist}(\mathbf{x}, \mathbf{x}') = 1$. Now, among all valid $\mathbf{f} \in \mathbb{Z}_+^E$ satisfying $\text{dist}(\partial_1 \mathbf{f}, \mathbf{x}') = 1$, choose $\mathbf{f}$ to minimize $\text{dist}(\partial_2 \mathbf{f}, \partial_2 \mathbf{h})$. If $\partial_1 \mathbf{f}$ is an $(\mathbf{x}', \mathbf{y})$ -step, then since $\mathbf{f}$ is valid we set $\mathbf{x}'' := \partial_1 \mathbf{f} \in \mathbb{J}_1$ and the two-step axiom is verified; otherwise, we know that $\mathbf{x}'$ is therefore an $(\partial_1 \mathbf{f}, \partial_1 \mathbf{h})$ -step by Proposition 3.2. Let $\mathbf{x}^{*} \in \mathbb{Z}_+^E$ be defined by $\mathbf{x}^{*}|_{V_1} = \mathbf{x}'$ and $\mathbf{x}^{*}|_{V_2} = \partial_2 \mathbf{f}$. Then $\mathbf{x}^{*}$ is an $(\partial \mathbf{f}, \partial \mathbf{h})$ -step, so by Lemma 4.2, we may choose $\mathbf{g} \in \mathbb{Z}_+^E$ so that $\partial \mathbf{g}$ is an $(\mathbf{x}^{*}, \partial \mathbf{h})$ -step. Now, depending on which side of $V(G)$ the one-unit move occurs, exactly one of the following cases is true:

- case 1. $\partial_1 \mathbf{g}$ is an $(\mathbf{x}', \mathbf{y})$ -step and $\partial_2 \mathbf{g} = \partial_2 \mathbf{f}$. This implies that $\mathbf{g}$ is valid since $\mathbf{f}$ is valid, so the two-step axiom is therefore verified by setting $\mathbf{x}'' := \partial_1 \mathbf{g}$ and we are done.
- case 2. $\partial_2 \mathbf{g}$ is an $(\partial_2 \mathbf{f}, \partial_2 \mathbf{h})$ -step and $\partial_1 \mathbf{g} = \mathbf{x}'$. Since $\mathbf{f}$ and $\mathbf{h}$ are valid, by the two-step axiom, at least one of the following statements has to be true:
 - (i). $\partial_2 \mathbf{g} \in \mathbb{J}_2$.
 - (ii). there exists an $(\partial_2 \mathbf{g}, \partial_2 \mathbf{h})$ -step, say $\mathbf{t} \in \mathbb{J}_2$.

Suppose (ii) holds. Then we may assume that $\mathbf{t} = \partial_2 \mathbf{g}'$ for some $\mathbf{g}' \in \mathbb{Z}_+^E$ where $\mathbf{g}' = \mathbf{g} \pm \mathbf{e}_e$ for some $e \in E$. Now we have that $\mathbf{g}'$ is valid and $\text{dist}(\partial_1 \mathbf{g}', \mathbf{x}') = \text{dist}(\partial_1 \mathbf{g}', \partial_1 \mathbf{g}) = 1$, but $\text{dist}(\partial_2 \mathbf{g}', \partial_2 \mathbf{h}) < \text{dist}(\partial_2 \mathbf{g}, \partial_2 \mathbf{h}) < \text{dist}(\partial_2 \mathbf{f}, \partial_2 \mathbf{h})$, contradicting our choice of $\mathbf{f}$. Hence, (i) must be true. This implies that $\mathbf{g}$ is valid, so $\mathbf{x}' = \partial_1 \mathbf{g} \in J_1$ and we are done.

Therefore, in either case, we get a desired vector in J_1 , and hence the proof is complete. $\square$

Remark 4.4. Fix an arbitrary bipartite graph $G = (V_1, V_2, E)$ and let $\mathbf{f} \in \mathbb{Z}_+^E$ be any vector on edges. Then

$$(\partial_1 \mathbf{f})_{V_1} = \mathbf{f}_E = (\partial_2 \mathbf{f})_{V_2}$$

It follows that for any constant-sum jump system $J_2 \subseteq \mathbb{Z}_+^{V_2}$, the associated jump system $\Theta_G(J_2) \subseteq \mathbb{Z}_+^{V_1}$ is also constant-sum.

Using our language and framework, Theorem 4.3 can naturally imply many results in the theory of jump systems.

Corollary 4.5. *Define a function $\phi : \mathbb{Z}^n \rightarrow \mathbb{Z}^{n-1}$ by*

$$(x_1, \dots, x_n) \longmapsto (x_1, \dots, x_{n-2}, x_{n-1} + x_n)$$

Then we have:

- A. If $J \subseteq \mathbb{Z}^n$ is a jump system, then $\phi(J)$ is a jump system.*
- B. If $K \subseteq \mathbb{Z}^{n-1}$ is a jump system, then $\phi^{-1}(K)$ is a jump system.*

Proof. Let $V_2 := [n]$ and $V_1 := [n-1]$. Construct a bipartite graph $G = (V_1, V_2, E)$ as follows: for each $i \in [n-2]$, join the vertex $i \in V_1$ to the vertex $i \in V_2$, and join the vertex $n-1 \in V_1$ to both vertices $n-1, n \in V_2$. That is, the graph looks like the identity matching on the first $n-2$ coordinates, while the last two vertices on V_2 are both identified with the last vertex on V_1 .

To see A, let $J \subseteq \mathbb{Z}_+^{V_2}$ be a jump system. We claim that

$$\Theta_G(J) = \phi(J)$$

Indeed, for any $\mathbf{f} = (f_1, \dots, f_{n-2}, f_{n-1}, f_n) \in \mathbb{Z}_+^E$ (with the obvious indexing), we have

$$\partial_2 \mathbf{f} = (f_1, \dots, f_{n-2}, f_{n-1}, f_n) \quad \text{and} \quad \partial_1 \mathbf{f} = (f_1, \dots, f_{n-2}, f_{n-1} + f_n)$$

Therefore, whenever $\partial_2 \mathbf{f} \in J$, we have $\partial_1 \mathbf{f} = \phi(\partial_2 \mathbf{f}) \in \phi(J)$. Conversely, for any $\mathbf{x} = (x_1, \dots, x_n) \in J$, define $\mathbf{f}$ by putting these entries on the corresponding edges. Then $\partial_2 \mathbf{f} = \mathbf{x}$ and $\partial_1 \mathbf{f} = \phi(\mathbf{x})$. Hence $\Theta_G(J) = \phi(J)$ and Theorem 4.3 tells us that $\phi(J)$ is a jump system.

To see B, we reverse the above construction: define the graph $G^{\text{op}} := (V_2, V_1, E)$. Let $K \subseteq \mathbb{Z}_+^{V_1}$ be a jump system. We claim that

$$\Theta_{G^{\text{op}}}(K) = \phi^{-1}(K)$$

Indeed, the reasoning is the same as above and Theorem 4.3 tells us that $\phi^{-1}(K)$ is a jump system. $\square$

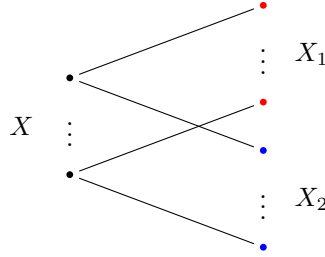
Remark 4.6. The assertion A is essentially [KS05, Theorem 4.2]. The assertion B follows from the same graphic viewpoint, and in our setting both assertions are immediate results of Theorem 4.3. We shall call the function ϕ a *merging* operation. Note that we can start from a jump system and do merging repeatedly; the results are all jump systems.

Corollary 4.7. *Let J_1 and J_2 be jump systems on the same ground set X . Then*

$$J_1 + J_2 := \{\mathbf{x} + \mathbf{y} \mid \mathbf{x} \in J_1 \text{ and } \mathbf{y} \in J_2\}$$

is a jump system.

Proof. We may assume that J_1, J_2 are non-negative by translation. Construct a bipartite graph $G = (V_1, V_2, E)$ as follows: Let V_1 represent X and $V_2 := X_1 \sqcup X_2$ where each X_i represents a copy of X . Let $E = E_1 \sqcup E_2$ where each E_i represents a perfect matching between X and X_i and each $e \in E$ joins two vertices representing the same element of X . The following is a sketch of the graph G :



Now, we claim that $\Theta_G(J_1 \oplus J_2) = J_1 + J_2$. To see this, take any $\mathbf{z} \in \Theta_G(J_1 \oplus J_2)$. Then, by definition, there exists a vector $\mathbf{f} \in \mathbb{Z}_+^E$ such that

$$\partial_1 \mathbf{f} = \mathbf{z} \quad \text{and} \quad \partial_2 \mathbf{f} \in J_1 \oplus J_2$$

Write $\partial_2 \mathbf{f} = \mathbf{x}_1 \oplus \mathbf{x}_2$ where $\mathbf{x}_i \in J_i$. By our construction $\mathbf{z} = \partial_1 \mathbf{f} = \mathbf{x}_1 + \mathbf{x}_2$, hence $\mathbf{z} \in J_1 + J_2$. Conversely, let $\mathbf{y} \in J_1 + J_2$. Then $\mathbf{y} = \mathbf{x}_1 + \mathbf{x}_2$ for some $\mathbf{x}_i \in J_i$. Note that $\mathbf{x}_1 \oplus \mathbf{x}_2 \in J_1 \oplus J_2$. Define $\mathbf{f} \in \mathbb{Z}_+^E$ by a copy of $\mathbf{x}_1 \oplus \mathbf{x}_2$ so that

$$\partial_1 \mathbf{f} = \mathbf{y} \quad \text{and} \quad \partial_2 \mathbf{f} = \mathbf{x}_1 \oplus \mathbf{x}_2$$

Thus, by definition $\mathbf{y} \in \Theta_G(J_1 \oplus J_2)$, as desired. Therefore, by Theorem 4.3, $J_1 + J_2$ is a jump system. $\square$

Remark 4.8. Using the same argument, we can generalize the result above to n jump systems. That is, let $J_1, \dots, J_n$ be jump systems on the same ground set X . Then

$$\Theta_G\left(\bigoplus_{i=1}^n J_i\right) = \sum_{i=1}^n J_i$$

where G is the bipartite graph whose left side is X , whose right side consists of n disjoint copies of X , and whose edges join corresponding elements. By this interpretation, we not only show that the sum of jump systems is also a jump system, but also obtain the structure of the summation through Rado-type theory. By restricting this associated jump system to the box $[\mathbf{0}, \mathbf{e}]$, we immediately recover the matroid union theorem.

Corollary 4.9. *Let $M_1, \dots, M_n$ be matroids on the same ground set X . Let*

$$\mathcal{I} := \left\{ \bigcup_{i=1}^n I_i \mid I_i \in \mathcal{I}(M_i), 1 \leq i \leq n \right\}$$

Then, $\mathcal{I}$ is the collection of independent sets of a matroid on X .

Definition 4.10. We call this matroid the *union* of $M_1, \dots, M_n$, denoted as $\bigvee_{i=1}^n M_i$.

Proof of Corollary 4.9. Note that equivalently, $I \in \mathcal{I}$ if and only if I can be partitioned into disjoint subsets $I_1, \dots, I_n$ such that $I_i \in \mathcal{I}_i$, $i \in [n]$. Now, construct the same bipartite graph $G(V_1, V_2, E)$ as in Remark 4.8. We encode every independent set in $\mathcal{I}_{M_i}$ by its indicating vector. Then, for each $i \in [n]$, $\mathbf{I}_i := \{\mathbf{1}_I \mid I \in \mathcal{I}_{M_i}\}$ is a jump system by Example 1.2. Let

$$J_2 := \bigoplus_{i=1}^n \mathbf{I}_i \quad \text{and} \quad J_1 := \Theta_G(J_2)$$

By Remark 4.8, the elements in J_1 are exactly the sums of independent vectors from $M_1, \dots, M_n$:

$$J_1 = \{\mathbf{1}_{I_1} + \dots + \mathbf{1}_{I_n} \mid I \in \mathbf{I}_i, i \in [n]\}$$

Now, let

$$J := J_1 \cap [\mathbf{0}, \mathbf{e}]$$

By Proposition 3.4 and Remark 3.5, J is a jump system. Evidently we have $J = \{\mathbf{1}_I \mid I \in \mathcal{I}\}$ by our construction of G . Note that J is down-closed and is inside the box $[\mathbf{0}, \mathbf{e}]$. Theorem 5.4 then tells us that $\mathcal{I}$ is the independent set of a matroid. $\square$

5. DISCRETE POLYMATROIDS AND JUMP SYSTEMS

We first state and prove a useful proposition: Given a discrete polymatroid P on the ground set V . For any $\mathbf{x} \in \mathbf{I}_P$, we say a subset $S \subseteq V$ is $\mathbf{x}$ -tight if $\mathbf{x}_S = \text{rk}_P(S)$. Then we claim:

Proposition 5.1. *Fix an arbitrary $\mathbf{x} \in \mathbf{I}_P$. Then, $\mathbf{x}$ -tight sets are closed under intersection and union.*

Proof. Let S and T be $\mathbf{x}$ -tight sets. Then, by submodularity, we have

$$\text{rk}_P(S \cap T) + \text{rk}_P(S \cup T) \geq \mathbf{x}_{S \cap T} + \mathbf{x}_{S \cup T} = \mathbf{x}_S + \mathbf{x}_T = \text{rk}_P(S) + \text{rk}_P(T) \geq \text{rk}_P(S \cap T) + \text{rk}_P(S \cup T)$$

This implies $\text{rk}_P(S \cap T) = \mathbf{x}_{S \cap T}$ and $\text{rk}_P(S \cup T) = \mathbf{x}_{S \cup T}$, as desired. $\square$

Now we prove our equivalence theorem 5.2.

Theorem 5.2. *For every $J \subseteq \mathbb{Z}_+^V$, the following are equivalent:*

- (i). $J = \mathbf{B}_P$ for a discrete polymatroid P .
- (ii). J is constant sum and for every $\mathbf{x}, \mathbf{y} \in J$ and $i \in V$ with $\mathbf{x}_i < \mathbf{y}_i$ there exists $j \in V$ with $\mathbf{x}_j > \mathbf{y}_j$ so that both $\mathbf{x} + \mathbf{e}_i - \mathbf{e}_j \in J$ and $\mathbf{y} - \mathbf{e}_i + \mathbf{e}_j \in J$.
- (iii). J is a constant sum jump system.
- (iv). J can be obtained from $\mathbf{B}_M$ for a matroid M by identification.

Proof of Theorem 5.2. We show that (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (iv) $\implies$ (i).

(i) $\implies$ (ii): J is clearly constant-sum by the definition of $\mathbf{B}_P$. Let $\mathbf{x}, \mathbf{y} \in J$ and assume that $\mathbf{x}_i < \mathbf{y}_i$. Observe that for every j with $\mathbf{x}_j > 0$,

$$\mathbf{x} + \mathbf{e}_i - \mathbf{e}_j \notin J \iff \exists \text{ a } \mathbf{x}\text{-tight set } A \subseteq V \text{ such that } i \in A \text{ but } j \notin A$$

and similarly, for every i with $\mathbf{y}_i > 0$,

$$\mathbf{y} - \mathbf{e}_i + \mathbf{e}_j \notin J \iff \exists \text{ a } \mathbf{y}\text{-tight set } B \subseteq V \text{ such that } i \notin B \text{ but } j \in B$$

By Proposition 5.1 there exists a unique minimal $\mathbf{x}$ -tight set, say S that contains i and a unique maximal $\mathbf{y}$ -tight set, say T that does not contain i . If there exists $j \in S \setminus T$ satisfying $\mathbf{x}_j > \mathbf{y}_j$, then by our above observation, both $\mathbf{x} + \mathbf{e}_i - \mathbf{e}_j$ and $\mathbf{y} - \mathbf{e}_i + \mathbf{e}_j$ are in J and we are done. Otherwise, note that

$$\mathbf{x}_S + \mathbf{y}_T = \text{rk}_P(S) + \text{rk}_P(T) \geq \text{rk}_P(S \cap T) + \text{rk}_P(S \cup T) \geq \mathbf{x}_{S \cap T} + \mathbf{y}_{S \cup T} = \mathbf{x}_S + \mathbf{y}_T + (\mathbf{y}_{S \setminus T} - \mathbf{x}_{S \setminus T})$$

But

$$\mathbf{y}_{S \setminus T} - \mathbf{x}_{S \setminus T} = \sum_{k \in S \setminus T} \mathbf{y}_k - \mathbf{x}_k = \mathbf{y}_i - \mathbf{x}_i + \sum_{k \in S \setminus i \setminus T} \mathbf{y}_k - \mathbf{x}_k > 0$$

Hence, we obtain a contradiction and the proof is done.

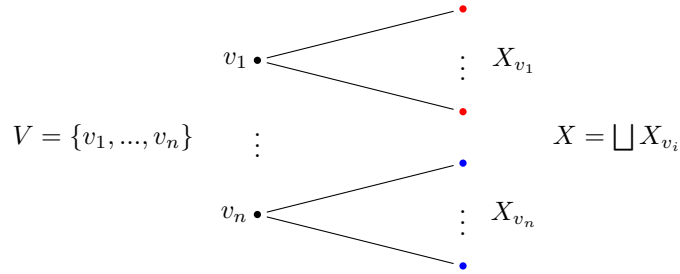
(ii) $\implies$ (iii): Let $\mathbf{x}, \mathbf{y} \in J$ and let $\mathbf{x}'$ be an arbitrary $(\mathbf{x}, \mathbf{y})$ -step. If $\mathbf{x}' = \mathbf{x} + \mathbf{e}_i$ for some $i \in V$, then $\mathbf{x}_i < \mathbf{y}_i$. By (ii), there exists $j \in V$ s.t. $\mathbf{x}_j > \mathbf{y}_j$ and $\mathbf{x} + \mathbf{e}_i - \mathbf{e}_j \in J$. Note that $\mathbf{x} + \mathbf{e}_i - \mathbf{e}_j$ is an $(\mathbf{x}', \mathbf{y})$ -step, so we are done; If $\mathbf{x}' = \mathbf{x} - \mathbf{e}_i$ for some $i \in V$, then (ii) implies that there exists $j \in V$ s.t. $\mathbf{y}_j > \mathbf{x}_j$ and

$\mathbf{x} - \mathbf{e}_i + \mathbf{e}_j \in J$. Note that $\mathbf{x} - \mathbf{e}_i + \mathbf{e}_j \in J$ is an $(\mathbf{x}', \mathbf{y})$ -step, so we are done. The two-step axiom is verified in both cases and the proof is complete.

(iii) $\implies$ (iv): Let $J \subseteq \mathbb{Z}_+^V$ be a constant-sum jump system with coordinate sum c . For each $v \in V$, take a set X_v of c new elements and let

$$X := \bigsqcup_{v \in V} X_v$$

Now construct a bipartite graph $G = (X, V, E)$ as follows: every element of X_v is adjacent only to v . Thus, each vertex on the right side X has degree one. By Theorem 4.3 we have the associated jump system $\tilde{J} := \Theta_G(J) \subseteq \mathbb{Z}_+^X$. Note that since J is constant-sum, Remark 4.4 implies that $\tilde{J}$ is also constant-sum.



Now define $\mathbf{B} := \tilde{J} \cap [\mathbf{0}, \mathbf{e}]$. It follows from our construction that $\mathbf{B} \neq \emptyset$. By Proposition 3.6 we know that $\mathbf{B} = \mathbf{B}_M$ for some matroid M . We claim that J is obtained from $\mathbf{B}_M$ by identification. For each $v \in V$, identify all elements of X_v into the single element v . Under this identification, a basis $B \in \mathcal{B}_M$ is sent to the vector $\mathbf{x} \in \mathbb{Z}_+^V$ where $\mathbf{x}_v = |B \cap X_v|$. By our construction of $\tilde{J}$, $\mathbf{x} \in J$. Conversely, if $\mathbf{x} \in J$, then $\mathbf{x}_V = c$, so $0 \leq \mathbf{x}_v \leq c$ for every $v \in V$. Since $|X_v| = c$, we may choose exactly $\mathbf{x}_v$ elements from X_v for each v , the resulting vector lies in $\mathbf{B}$ and its identification is exactly $\mathbf{x}$. In other words, by defining the reversed graph $G^{\text{op}} := (V, X, E)$, we have

$$J = \Theta_{G^{\text{op}}}(\Theta_G(J) \cap [\mathbf{0}, \mathbf{e}])$$

(iv) $\implies$ (i): Suppose that J is obtained from $\mathbf{B}_M$ by identification. That is, there exists a matroid M on a finite set $X = \bigsqcup_{v \in V} X_v$ such that

$$J = \{\mathbf{x} \in \mathbb{Z}_+^V \mid \text{there exists } B \in \mathcal{B}_M \text{ such that } \mathbf{x}_v = |B \cap X_v| \text{ for every } v \in V\}$$

Write $X_S := \bigsqcup_{v \in S} X_v$ and define $f : 2^V \rightarrow \mathbb{Z}_+$ by

$$f(S) := \text{rk}_M(X_S)$$

We claim that f is the rank function for a discrete polymatroid P on V . Indeed, the normalization and the monotonicity are evident. To see the submodularity, note that

$$\begin{aligned} f(S) + f(T) &= \text{rk}_M(X_S) + \text{rk}_M(X_T) \\ &\geq \text{rk}_M(X_S \cap X_T) + \text{rk}_M(X_S \cup X_T) = \text{rk}_M(X_{S \cap T}) + \text{rk}_M(X_{S \cup T}) = f(S \cap T) + f(S \cup T) \end{aligned}$$

Now, we show that $J = \mathbf{B}_P$. Take an arbitrary $\mathbf{x} \in J$. Then there exists $B \in \mathcal{B}_M$ such that $\mathbf{x}_v = |B \cap X_v|$ for every $v \in V$. So for every $S \subseteq V$ we have

$$\mathbf{x}_S = |B \cap X_S| \leq \text{rk}_M(X_S) = \text{rk}_P(S)$$

Also

$$\mathbf{x}_V = |B| = \text{rk}_M(X) = \text{rk}_P(V)$$

Hence $\mathbf{x} \in \mathbf{B}_P$ and $J \subseteq \mathbf{B}_P$. Conversely, take an arbitrary $\mathbf{x} \in \mathbf{B}_P$. We need to find a basis $B \in \mathcal{B}_M$ such that $|B \cap X_v| = \mathbf{x}_v$ for every $v \in V$. For every $v \in V$ let Y_v be a set of $\mathbf{x}_v$ new elements. Let $Y := \bigsqcup Y_v$.

Construct a bipartite graph between Y and X by joining each copy of v to every element of X_v . For each $A \subseteq Y$, let

$$S_A := \{v \in V : A \text{ contains at least one copy of } v\}$$

Then $N(A) = X_{S_A}$. Moreover, since A contains at most $\mathbf{x}_v$ copies of each v , we have $|A| \leq \mathbf{x}_{S_A}$. Hence

$$|A| \leq \mathbf{x}_{S_A} \leq \text{rk}_P(S_A) = \text{rk}_M(X_{S_A}) = \text{rk}_M(N(A))$$

Therefore, by Rado's theorem for matroids 1.10, there exists a matching between Y and an independent set, say $B \in \mathcal{I}_M$. Now

$$|B| = |Y| = \mathbf{x}_V = \text{rk}_P(V) = \text{rk}_M(X)$$

so B is a basis of M . Moreover, since the $\mathbf{x}_v$ copies of v are matched into X_v , we have $|B \cap X_v| = \mathbf{x}_v$ for every $v \in V$, as desired. So $\mathbf{x} \in J$ and the proof is complete. $\square$

Before we prove Theorem 5.4, we first state and prove a useful lemma:

Lemma 5.3. *Let P be a discrete polymatroid. Then $\mathbf{I}_P = \downarrow \mathbf{B}_P$.*

Proof. Every vector below a vector in $\mathbf{B}_P$ clearly belongs to $\mathbf{I}_P$. Conversely, take an arbitrary $\mathbf{x} \in \mathbf{I}_P$. Suppose that $\mathbf{x}_V < \text{rk}_P(V)$ (i.e., $\mathbf{x} \notin \mathbf{B}_P$). Then we can increase a coordinate of $\mathbf{x}$ by one and still remain inside $\mathbf{I}_P$. Because if not, then for every $v \in V$, there exists an $\mathbf{x}$ -tight set containing v . Since $\mathbf{x}$ -tight sets are closed under union (Proposition 5.1), their union V is also $\mathbf{x}$ -tight. Hence $\mathbf{x}_V = \text{rk}_P(V)$, a contradiction. Therefore, starting from $\mathbf{x}$, we may repeatedly increase coordinates while staying inside $\mathbf{I}_P$ until we reach some vector $\mathbf{y} \in \mathbf{B}_P$ with $\mathbf{x} \preceq \mathbf{y}$. Hence, $\mathbf{x} \in \downarrow \mathbf{B}_P$, as desired. $\square$

Theorem 5.4. *For every $J \subseteq \mathbb{Z}_+^V$, the following are equivalent:*

- (i). $J = \mathbf{I}_P$ for a discrete polymatroid P .
- (ii). J is a down-closed jump system.
- (iii). J can be obtained from $\mathbf{I}_M$ for a matroid M by identification.

Proof of Theorem 5.4. We show that (i) $\implies$ (iii) $\implies$ (ii) $\implies$ (i).

(i) $\implies$ (iii): By Lemma 5.3 we know that $J = \mathbf{I}_P = \downarrow \mathbf{B}_P$. By Theorem 5.2, $\mathbf{B}_P$ can be obtained from $\mathbf{B}_M$ for a matroid M by identification. Under the same identification, the down-closure of $\mathbf{B}_M$ is sent exactly to the down-closure of $\mathbf{B}_P$. But $\downarrow \mathbf{B}_M = \mathbf{I}_M$, hence $\mathbf{I}_P$ can be obtained from $\mathbf{I}_M$ by this identification.

(iii) $\implies$ (ii): Since $\mathbf{I}_M$ is a jump system and identification is a special case of the operation Θ , Theorem 4.3 implies that J is a jump system. To see J is down-closed, take an arbitrary $\mathbf{x} \in J$ and let $\mathbf{z} \preceq \mathbf{x}$. Since $\mathbf{x}$ is obtained from $\mathbf{1}_I \in \mathbf{I}_M$ for some independent set I , we may choose a subset $I' \subseteq I$ whose image is exactly $\mathbf{z}$. Since $\mathbf{I}_M$ is hereditary, $\mathbf{1}_{I'} \in \mathbf{I}_M$. Hence $\mathbf{z} \in J$, as desired.

(ii) $\implies$ (i): By Proposition 3.7, $J_{\max}$ is a constant-sum jump system. Hence, by Theorem 5.2, there exists a discrete polymatroid P such that $J_{\max} = \mathbf{B}_P$. We claim that $J = \downarrow J_{\max}$. Indeed, $\downarrow J_{\max} \subseteq J$ follows immediately from the down-closedness of J . Conversely, take an arbitrary $\mathbf{x} \in J$. By Proposition 3.8, there exists $\mathbf{y} \in J_{\max}$ such that $\mathbf{x} \preceq \mathbf{y}$. Hence, $\mathbf{x} \in \downarrow J_{\max}$, as desired. Therefore, $J = \downarrow J_{\max} = \downarrow \mathbf{B}_P = \mathbf{I}_P$ by Lemma 5.3 and the proof is done. $\square$

6. THE KÖNIG-TYPE THEOREM AND THE RADO-TYPE RANK FUNCTION FORMULA FOR DISCRETE POLYMATROIDS

Lemma 6.1. *Let $G = (V_1, V_2, E)$ be a bipartite graph, and let M_1, M_2 be matroids on V_1, V_2 , respectively. Then*

$$\max_{\substack{M \text{ is a matching} \\ V(M) \cap V_1 \in \mathcal{I}_{M_1} \\ V(M) \cap V_2 \in \mathcal{I}_{M_2}}} \{|M|\} = \min_{\substack{X_1 \subseteq V_1 \\ X_2 \subseteq V_2 \\ X_1 \sqcup X_2 \text{ is a vertex cover}}} \{\text{rk}_{M_1}(X_1) + \text{rk}_{M_2}(X_2)\}$$

Proof. This is a direct corollary of Theorem 1.17 and Theorem 1.10. $\square$

Theorem 6.2 (König-type Theorem for Discrete Polymatroids). *Let $G = (V_1, V_2, E)$ be a bipartite graph, and let P_1, P_2 be discrete polymatroids on V_1, V_2 , respectively. Then*

$$\max_{\substack{\partial_1 \mathbf{f} \in \mathbf{I}_{P_1} \\ \partial_2 \mathbf{f} \in \mathbf{I}_{P_2}}} \{\mathbf{f}_E\} = \min_{\substack{X_1 \subseteq V_1 \\ X_2 \subseteq V_2 \\ X_1 \sqcup X_2 \text{ is a vertex cover}}} \{\text{rk}_{P_1}(X_1) + \text{rk}_{P_2}(X_2)\}$$

Proof. By Theorem 5.4, for each $i = 1, 2$, we may choose a matroid M_i on a finite set

$$U_i = \bigsqcup_{v \in V_i} U_{i,v}$$

such that $\mathbf{I}_{P_i}$ is obtained from $\mathbf{I}_{M_i}$ by identifying all elements of $U_{i,v}$ with v . Thus, for every $X_i \subseteq V_i$, we have

$$\text{rk}_{P_i}(X_i) = \text{rk}_{M_i} \left(\bigsqcup_{v \in X_i} U_{i,v} \right)$$

Now, construct a bipartite graph $G^* = (U_1, U_2, E^*)$ as follows: for every edge $uv \in E$, with $u \in V_1$ and $v \in V_2$, put a complete bipartite graph between $U_{1,u}$ and $U_{2,v}$. We claim that the left-hand side of the theorem is equal to

$$\max_{\substack{M \text{ is a matching of } G^* \\ V(M) \cap U_1 \in \mathcal{I}_{M_1} \\ V(M) \cap U_2 \in \mathcal{I}_{M_2}}} \{|M|\}$$

Indeed, a matching M in G^* gives a vector $\mathbf{f} \in \mathbb{Z}_+^E$ by letting $\mathbf{f}_{uv}$ be the number of matching edges between $U_{1,u}$ and $U_{2,v}$. Under the identification, the two vertex sets of M become exactly $\partial_1 \mathbf{f}$ and $\partial_2 \mathbf{f}$. Conversely, any $\mathbf{f} \in \mathbb{Z}_+^E$ with $\partial_i \mathbf{f} \in \mathbf{I}_{P_i}$ can be lifted to such a matching in G^* , since the graph between each $U_{1,u}$ and $U_{2,v}$ is complete bipartite. In both directions, the size is preserved: $|M| = \mathbf{f}_E$.

Hence, by Lemma 6.1, the left-hand side of the theorem is equal to

$$\min_{\substack{C_1 \subseteq U_1 \\ C_2 \subseteq U_2 \\ C_1 \sqcup C_2 \text{ is a vertex cover of } G^*}} \{\text{rk}_{M_1}(C_1) + \text{rk}_{M_2}(C_2)\}$$

It remains to translate this minimum back to G . If $X_1 \sqcup X_2$ is a vertex cover of G , then

$$\left(\bigsqcup_{u \in X_1} U_{1,u} \right) \sqcup \left(\bigsqcup_{v \in X_2} U_{2,v} \right)$$

is a vertex cover of G^* , and its rank contribution is $\text{rk}_{P_1}(X_1) + \text{rk}_{P_2}(X_2)$. Conversely, let $C_1 \sqcup C_2$ be a vertex cover of G^* . Define

$$X_i := \{v \in V_i : U_{i,v} \subseteq C_i\}$$

Since every edge $uv \in E$ gives a complete bipartite graph between $U_{1,u}$ and $U_{2,v}$, the cover property forces either $U_{1,u} \subseteq C_1$ or $U_{2,v} \subseteq C_2$. Hence $X_1 \sqcup X_2$ is a vertex cover of G . Moreover,

$$\text{rk}_{P_i}(X_i) = \text{rk}_{M_i} \left(\bigsqcup_{v \in X_i} U_{i,v} \right) \leq \text{rk}_{M_i}(C_i)$$

Therefore, passing from $C_1 \sqcup C_2$ to $X_1 \sqcup X_2$ cannot increase the value. Thus the two minima are equal, and the theorem follows. $\square$

Theorem 6.3 (Rado-type Rank Function Formula). *Let $G = (V_1, V_2, E)$ be a bipartite graph, P_2 be a discrete polymatroid on V_2 , and let $\mathbf{c} \in \mathbb{Z}_+^{V_1}$. Then*

$$J := \{\mathbf{x} \in \mathbb{Z}_+^{V_1} : \mathbf{x} \preceq \mathbf{c}, \mathbf{x} = \partial_1 \mathbf{f}, \partial_2 \mathbf{f} \in \mathbf{I}_{P_2} \text{ for some } \mathbf{f} \in \mathbb{Z}_+^E\}$$

is the independent set $\mathbf{I}_{P_1}$ of a discrete polymatroid P_1 on V_1 , with cage $\mathbf{c}$. Moreover, the rank function of P_1 is given by

$$\text{rk}_{P_1}(S) = \min_{T \subseteq S} \{\text{rk}_{P_2}(N(T)) + \mathbf{c}_{S \setminus T}\}$$

for every $S \subseteq V_1$.

Proof. Note that in our language, we have

$$J = \Theta_G(\mathbf{I}_{P_2}) \cap [\mathbf{0}, \mathbf{c}]$$

By Theorem 5.4, $\mathbf{I}_{P_2}$ is a down-closed jump system. Hence Theorem 4.3 tells us that $\Theta_G(\mathbf{I}_{P_2})$ is a jump system, and by Proposition 3.4 and Remark 3.5, J is also a jump system. Moreover, J is evidently down-closed. Thus, by Theorem 5.4, there exists a discrete polymatroid P_1 on V_1 such that $J = \mathbf{I}_{P_1}$. Since every vector in J is bounded above by $\mathbf{c}$, we have $\text{rk}_{P_1}(\{v\}) \leq \mathbf{c}_v$ for every $v \in V_1$. Hence P_1 has cage $\mathbf{c}$. It remains to compute the rank function of P_1 . Fix $S \subseteq V_1$. Recall that

$$\text{rk}_{P_1}(S) = \max\{\mathbf{x}_S \mid \mathbf{x} \in \mathbf{I}_{P_1}\}$$

Since $\mathbf{I}_{P_1} = J$, this is equal to the maximum of $\mathbf{f}_E$ over the bipartite graph induced by S and V_2 , denoted as $G[S, V_2]$, subject to the conditions

$$\partial_1 \mathbf{f} \preceq \mathbf{c}|_S \quad \text{and} \quad \partial_2 \mathbf{f} \in \mathbf{I}_{P_2}$$

Apply Theorem 6.2 to this induced bipartite graph, together with the discrete polymatroid on S whose independent vectors are exactly the vectors bounded above by $\mathbf{c}|_S$. Its rank function is $A \mapsto \mathbf{c}_A$. We obtain

$$\text{rk}_{P_1}(S) = \min_{\substack{A \subseteq S \\ B \subseteq V_2 \\ A \sqcup B \text{ is a vertex cover of } G[S, V_2]}} \{\mathbf{c}_A + \text{rk}_{P_2}(B)\}$$

For a fixed $A \subseteq S$, the smallest possible choice of B is $N(S \setminus A)$, since B must cover all edges leaving $S \setminus A$. Therefore

$$\text{rk}_{P_1}(S) = \min_{A \subseteq S} \{\mathbf{c}_A + \text{rk}_{P_2}(N(S \setminus A))\}$$

Replacing $S \setminus A$ by T , we obtain

$$\text{rk}_{P_1}(S) = \min_{T \subseteq S} \{\text{rk}_{P_2}(N(T)) + \mathbf{c}_{S \setminus T}\}$$

as desired. □

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